

KYUBIN JAMES KWON

[NASA/ADS](#)

RESEARCH INTEREST

Tidal interactions, neutron stars, gravitational waves, statistical methods, asteroseismology

EDUCATION

University of California, Santa Barbara 2021–2026 (expected)

Ph.D. Physics (expected)

M.A. Physics

2024

Advisor: Prof. Tejaswi Venumadhav

University of California, Berkeley

2021

High Distinction, B.A. Honors Astrophysics; B.A. Physics

Thesis: *deepCR on ACS/WFC: Cosmic-Ray Rejection for HST ACS/WFC Photometry*

PUBLICATION

1. Huang, X.; ..., **Kwon, K.J.** *et al.* 2025, arXiv e-prints, arXiv:2509.18089, doi:[10.48550/arXiv.2509.18089](#), *DESI Strong Lens Foundry II: DESI Spectroscopy for Strong Lens Candidates*
2. Hegade, A., **Kwon, K.J.** *et al.* 2025, arXiv e-prints, arXiv:2507.10693, doi: [10.48550/arXiv.2507.10693](#), *Relativistic and Dynamical Love*
3. **Kwon, K.J.**, Yu, H., & Venumadhav, T. 2025, arXiv e-prints, arXiv:2503.11837, doi: [10.48550/arXiv.2503.11837](#), *Resonance locking: radian-level phase shifts due to nonlinear hydrodynamics of g-modes in merging neutron star binaries*
4. **Kwon, K.J.**, Yu, H., & Venumadhav, T. 2024, arXiv e-prints, arXiv:2410.03831, doi: [10.48550/arXiv.2410.03831](#), *Resonance Locking of Anharmonic g-Modes in Coalescing Neutron Star Binaries*
5. Huang, X.; ..., **Kwon, K.J.** *et al.* 2025, arXiv e-prints, arXiv:2502.03455, doi:[10.48550/arXiv.2502.03455](#), *DESI Strong Lens Foundry I: HST Observations and Modeling with GIGA-Lens*
6. **Kwon, K.J.**, & Hahn, C. 2024, ApJ, 976, 76, doi: [10.3847/1538-4357/ad8442](#), *Investigating the Kinematics of Central and Satellite Galaxies Using Normalizing Flow*
7. Lovell, C.C., ..., **Kwon, J.**, ..., *et al.* 2023, in Machine Learning for Astrophysics, 21, doi: [10.48550/arXiv.2307.06967](#), *A Hierarchy of Normalizing Flows for Modelling the Galaxy-Halo Relationship*
8. Yu, H., Weinberg, N.N., Arras, P., **Kwon, J.**, & Venumadhav, T. 2023, Monthly Notices of the Royal Astronomical Society, 519, 4325, doi: [10.1093/mnras/stac3614](#), *Beyond the linear tide: impact of the non-linear tidal response of neutron stars on gravitational waveforms from binary inspirals*
9. Storfer, C., ..., **Kwon, J.**, ..., *et al.* 2024, ApJS, 274, 16, doi: [10.3847/1538-4365/ad527e](#), *New Strong Gravitational Lenses from the DESI Legacy Imaging Surveys Data Release 9*
10. **Kwon, K.J.**, Hahn, C., & Alsing, J. 2023, ApJS, 265, 23, doi: [10.3847/1538-4365/acba14](#), *Neural Stellar Population Synthesis Emulator for the DESI PROVABGS*
11. Hahn, C., **Kwon, K.J.**, *et al.* 2023, ApJ, 945, 16, doi: [10.3847/1538-4357/ac8983](#), *The DESI PRObabilistic Value-added Bright Galaxy Survey (PROVABGS) Mock Challenge*
12. **Kwon, K.J.**, Zhang, K., & Bloom, J.S. 2021b, RNAAS, 5, 98, doi: [10.3847/2515-5172/abf6c8](#), *deepCR on ACS/WFC: Cosmic-Ray Rejection for HST ACS/WFC Photometry*
13. **Kwon, K.J.**, Zhang, K., & Bloom, J.S. 2021a, in AAS Meeting Abstracts, Vol. 237, AAS Meeting Abstracts, 350.02, *deepCR-ACS/WFC: Cosmic-Ray Rejection Model for HST ACS/WFC Photometry*

ACADEMIC/RESEARCH EXPERIENCE

Referee, ApJS	2026
Academic Visitor, <i>University of Birmingham, United Kingdom</i>	2025
Academic Visitor, <i>International Centre for Theoretical Sciences, India</i>	2023, 2024
Graduate Student Researcher, <i>UC Santa Barbara, USA</i>	2022–
Research Intern, <i>NASA Ames & KARI¹, USA</i>	Postponed (Ames Center closed due to COVID-19)
Undergraduate Research Apprentice Program, <i>UC Berkeley, USA</i>	2019–2021

COMPUTING EXPERIENCE

Languages	Python, C++, Shell scripting
Methods	High-Performance Computing (HPC) Parallel Computing (MPI, OpenMP) Machine learning (TensorFlow, PyTorch)
HPC Machine	UCSB HPC Cluster National Energy Research Scientific Computing Center–Cori (2M+ CPU hours) Exalearn (Lawrence Berkeley National Laboratory)

TEACHING EXPERIENCE

Course Reader (Statistical Mechanics), UCSB	2026
Instructional Laboratory Group, UCSB	2025
Lead Teaching Assistant (Physics Laboratory), UCSB	2022
Teaching Assistant (Physics Laboratory), UCSB	2021
Course Reader (Relativistic Astrophysics and Cosmology), UC Berkeley	2021
Course Reader (Stellar Physics), UC Berkeley	2020
Math Tutor, Diablo Valley College	2018-2019

AWARDS, CERTIFICATES, ETC.

Academic Senate Doctoral Student Travel Grant, UCSB	2026
DGRAV Student Travel Award, APS	
Academic Senate Doctoral Student Travel Grant, UCSB	2025
Graduate Student Association Travel Grant, UCSB	
Phi Beta Kappa, UC Berkeley	2021
Chambliss Astronomy Award Honorable Mention, AAS 237	
Dean’s List-College of Letters & Science, UC Berkeley	

TALKS & WRITINGS

APS Global Physics Summit, <i>Resonance Locking of g-Modes in Coalescing Neutron Star Binaries</i>	2026
APS Global Physics Summit, <i>General Relativistic Triplet Interaction of Oscillation Eigenmodes in Deformed Neutron Stars</i>	
GWPAW25, <i>Resonance Locking of g-Modes in Coalescing Neutron Star Binaries</i>	2025
University of Birmingham Astrophysics Seminar*, <i>Nonlinear Tides in Coalescing BNS</i>	
The 41 th Pacific Coast Gravity Meeting, <i>Resonance Locking of g-Modes in Neutron Star Binaries</i>	
APS Global Physics Summit, <i>Resonance Locking of g-Modes in Coalescing Neutron Star Binaries</i>	
UCSB AstroLunch, <i>Resonance Locking of g-Modes in Coalescing Neutron Star Binaries</i>	2024
ICTS Astrophysics & Relativity Seminar*, <i>Nonlinear Tides in Coalescing Neutron Star Binaries</i>	
GWPAW24, <i>Nonlinear Resonance Locking in Coalescing Binary Neutron Stars</i>	

¹Korea Aerospace Research Institute

Montana State University Gravity Group Seminar*, <i>Nonlinear Tidal Effects on BNS Inspiral</i>	
SBI for Galaxy Evolution, <i>Normalizing Flow to Model the Kinematics of Galaxies in Halos</i>	
The 40 th Pacific Coast Gravity Meeting, <i>Tidal pg-instability in Coalescing Binary Neutron Stars</i>	
UCSB AstroTea, <i>Neural Stellar Population Synthesis Emulator for Probabilistic VAC</i>	2023
UCSB Astro Lunch, <i>The Probabilistic Value-Added Catalog for Bright Galaxy Survey</i>	2022
The 54th Sierra Conference , <i>The Probabilistic Value-Added BGS Catalog</i>	
Astrobites *, Invited Writer	2021
AAS237 , <i>deepCR on ACS/WFC: Cosmic-Ray Rejection Model for HST ACS/WFC Photometry</i>	
DESI Research Forum, <i>GQP Mock Challenge for Probabilistic Value-Added BGS Catalog</i>	2020

*: Invited

ACADEMIC SERVICE & VOLUNTEER

UCSB AstroTea Organizer	2023–
Language Mentor for War-Impacted Ukrainian Students	2023–
Co-chair of Astrophysics Departmental Undergraduate Task Force, UC Berkeley	2020–2021
Astrophysics Finals Workshop Facilitator	
Astrophysics Launch Day Orientation Facilitator	
Transfer Exploration Day Panel	
Town Hall Astro Social Panel	
Astro Summer Social Panel	

REFERENCES

Tejaswi Venumadhav
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